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Module 6 Introduction

Lesson 1: Numerical Methods and Approximations in Bayesian Computation

Lesson 2: Simulation Techniques for Bayesian Inference

Lesson 3: Advanced Markov Chain Methods for Bayesian Computation

Module 6 Summative Assessment

Graded Assignment: Module 6 Summative Assessment Submitted

Module 6 Summary

Your grade: 93.33%

Your latest: **93.33%** • Your highest: **93.33%** • To pass you need at least 80%. We keep your highest score.

Next item

Module 6 Summative Assessment

Review Learning Objectives

Instructions

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

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Assignment details

Due Attempts

Submitted
Dec 9, 2:25 AM IST

Retry

Your grade
To pass you need at least 80%. We keep your highest score.

93.33%

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1. Why might we use importance resampling? **1 / 1 point**

- To generate dependent samples
- To simplify the posterior distribution
- To convert weighted samples into unweighted samples
- To avoid the use of MCMC methods

Correct
Correct! Importance resampling converts weighted samples into unweighted samples, facilitating further analysis.

2. Why is it useful to compare crude inferences with later, more detailed results? **1 / 1 point**

- To avoid computational errors
- To ensure the model is not too complex
- To validate the more detailed results
- To simplify the model

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Correct
Correct! Comparing crude inferences with more detailed results helps validate the more detailed analysis.

3. Based on the reading, what is an ideal property of a Metropolis-Hastings jumping distribution? **1 / 1 point**

- It should be difficult to sample from
- It should be symmetric
- It should always lead to independent samples
- It should have a high acceptance rate

Correct
Correct! A good jumping distribution should not be rejected too frequently, ensuring a reasonable acceptance rate.

4. Based on the reading, what is the purpose of using fake data in model debugging? **1 / 1 point**

- To simplify the posterior distribution
- To avoid convergence issues
- To simulate real-world data
- To check if the model produces reasonable inferences

Correct
Correct! Fake data is used to check if the model produces reasonable inferences.

5. What is the basic idea of the Gibbs sampler? **1 / 1 point**

- To simplify the posterior distribution
- To sample each parameter independently
- To sample from the joint distribution directly
- To sample each parameter conditional on the other parameters

Correct
Correct! The Gibbs sampler samples each parameter conditional on the other parameters.

6. What is the simplest deterministic method for numerical integration? **1 / 1 point**

- Importance sampling
- Monte Carlo integration
- Rejection sampling
- Grid-based integration

Correct
Correct! The simplest deterministic method is grid-based integration.

7. What is rejection sampling? **1 / 1 point**

- A method that avoids using a target distribution
- A method to draw samples from an easy-to-sample distribution and reject some to match the target distribution
- A method to ensure all samples are accepted
- A method that always improves computational efficiency

Correct
Correct! Rejection sampling involves drawing samples from a proposal distribution and accepting or rejecting them to match the target distribution.

8. Suppose we wish to estimate the integral $\int sin\theta cos\theta d\theta$ from 0 to $\sqrt{2}$. Which of the following correspond to the trapezoidal rule? Assume $S = 5, \theta^1 = 0, \theta^5 = \sqrt{2}$, and θ^i are equidistant. **1 / 1 point**

- $p\left(\theta|y\right) = cos\left(\theta\right), h\left(\theta\right) = sin\left(\theta\right), w_0 = \sqrt{2}, w_5 = \sqrt{2}$, and $w_s = 2\sqrt{2}$
- $p\left(\theta|y\right) = 1/\sqrt{2}, h\left(\theta\right) = \sqrt{2}sin\left(\theta\right)cos\left(\theta\right), w_0 = \sqrt{2}/8, w_5 = \sqrt{2}/84$, and $w_s = \sqrt{2}/4$ for all other values of s
- None of the above
- $p\left(\theta|y\right) = sin\left(\theta\right)cos\left(\theta\right), h\left(\theta\right) = 1, w_0 = 1, w_5 = 1$, and $w_s = 2$ for all other values of s

Correct
Correct! We have a probability distribution for $p\left(\theta|y\right)$. That, combined with the weights yields the approximating corresponding to the trapezoidal rule.

9. Why are logarithms of posterior densities often used in Bayesian computation? **1 / 1 point**

- To avoid computational overflows
- To better correspond with the maximum likelihood estimate
- To linearize the data
- To simplify the computation of integrals

Correct
Correct! Using logarithms of posterior densities helps avoid computational overflows.

10. What is the Metropolis algorithm based on? **1 / 1 point**

- A random walk with an acceptance/rejection rule
- Sampling directly from the posterior distribution
- Deterministic sampling
- A fixed step size

Correct
Correct! The Metropolis algorithm is based on a random walk with an acceptance/rejection rule to converge to the target distribution.

11. According to the reading and your experience, what is a major advantage of using a general Bayesian software package like Stan or Bugs? **1 / 1 point**

- They eliminate the need for model checking
- They guarantee faster computation than custom code
- They allow users to fit Bayesian models without deep mathematical or computational knowledge
- They can handle all possible Bayesian models

Correct
Correct! These packages enable users to fit Bayesian models without needing extensive mathematical or computational expertise.

12. What is an advantage of direct simulation in Bayesian models? **1 / 1 point**

- It always provides exact posterior distributions
- It requires fewer computational resources than other methods
- It does not need conjugate priors
- It is often easy to draw from the posterior distribution directly

Correct
Correct! In simple nonhierarchical Bayesian models, it is often easy to draw from the posterior distribution directly.

13. What is a potential issue with Markov chain simulation? **0 / 1 point**

- It requires a large number of starting points
- It always converges quickly
- It may produce dependent samples
- It eliminates the need for prior distributions

Incorrect
While multiple starting points are useful, they are not always required.

14. Which of the following is true about most simulation methods used in Bayesian computation? **1 / 1 point**

- They provide exact values of integration
- They rely on random samples
- They are analytically derived
- They are deterministic algorithms

Correct
Correct! Simulation methods rely on obtaining random samples from the desired distribution.

15. What is a key difference between the Metropolis and Metropolis-Hastings algorithms? **1 / 1 point**

- The Metropolis-Hastings algorithm can use asymmetric jumping rules
- The Metropolis algorithm always rejects samples
- The Metropolis algorithm uses asymmetric jumping rules
- The Metropolis-Hastings algorithm requires conjugate priors

Correct
Correct! The Metropolis-Hastings algorithm generalizes the Metropolis algorithm by allowing asymmetric jumping rules.